

t-Confidence Intervals Practice

MATH 213

Group 1

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

In the study itself, the scientists compared the three groups of mice. Here, we will examine one of the groups, separately, for now.

1. In the dim light group, there were 10 mice. After eight weeks, they gained an average of 7.9g with a standard deviation of 3.0g. Calculate a 95% confidence interval for the **mean weight** gain of mice exposed to dim light at night. (Here, we are assuming that the mice in this group are representative of all mice.) Show your work. If you use R, tell what commands you use.

```
t_star = qt(p=0.995, df=9)
cf_low <- 7.9 - t_star * (3/sqrt(10))
cf_high <- 7.9 + t_star * (3/sqrt(10))
```

(5.75, 10.05)

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night.

```
t_star = qt(p=0.995, df=9)
cf_low <- 7.9 - t_star * (3/sqrt(10))
cf_high <- 7.9 + t_star * (3/sqrt(10))
```

(4.82, 10.83)

Group 2

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95%:

```
## {r}  
7.9 + 2.262157 * (3.0/sqrt(10))
```

```
[1] 10.04607
```

```
## {r}  
7.9 - 2.262157 * (3.0/sqrt(10))
```

```
[1] 5.753929
```

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

99%:

```
## {r}  
qt(p = 0.995, df=9)
```

```
[1] 3.249836
```

```
## {r}  
7.9 + 3.249836 * (3.0/sqrt(10))  
##
```

```
[1] 10.98307
```

```
## {r}  
7.9 - 3.249836 * (3.0/sqrt(10))  
##
```

```
[1] 4.816935
```

Group 3

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

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`qt(0.975, df = n - 1)`

5.753929

10.046071

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

$t(0.995, df = n - 1)$

4.816935 10.983065

Group 4

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

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```
```{r}
7.9+qt(p = 0.975, df = 9)*(3/sqrt(10))
```
```

```
[1] 10.04607
```

```
```{r}
7.9-qt(p = 0.975, df = 9)*(3/sqrt(10))
```
```

```
[1] 5.753929
```


2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

```
## {r}  
7.9+qt(p = .995, df = 9)*(3/sqrt(10))  
##
```

```
[1] 10.98306
```

```
## {r}  
7.9-qt(p = .995, df = 9)*(3/sqrt(10))  
##
```

```
[1] 4.816935
```

Group 5

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

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$$df = n - 1 = 10 - 1 = 9$$

$$t\text{-score} = 2.262157$$

$$se = 3 / \sqrt{10} = .949$$

$$2.262157 * .949 = 2.146786993$$

$$7.9 - 2.146786993 = 5.753213007 \quad 7.9 + 2.146786993 = 10.046786993$$

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night. .

$$t_{score}=3.249836$$

$$3.249836*.949=3.084094364$$

$$7.9-3.084094364=4.815905636$$

$$7.9+3.084094364= 10.984094364$$

Group 6

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

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```
t_star <- (qt(p = 0.975, df = 9))  
  
t_star  
[1] 2.262157  
  
{r}  
7.9 + t_star * (3/sqrt(10))  
7.9 - t_star * (3/sqrt(10))  
  
[1] 10.04607  
[1] 5.753929
```

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

```
qt(p = .995, df = 9)
```

```
```\r\nt_star_2 <- (qt(p = 0.995, df = 9))\n\nt_star_2
```

```
[1] 3.249836
```

```
```\r\n7.9 + t_star_2 * (3/sqrt(10))\n7.9 - t_star_2 * (3/sqrt(10))\n```\r\n
```

```
[1] 10.98306\n[1] 4.816935
```

Group 7

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

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$$\bar{X} \pm t^*(s/\sqrt{n})$$

$$qt(p = 0.975, df = 9)$$

$$T^* = 2.26$$

$$7.9 \pm 2.26*(3/\sqrt{10})$$

$$(5.76, 10.04)$$

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

$\bar{X} \pm t^*(s/\sqrt{n})$

$qt(p = 0.99, df = 9)$

3.249836

$7.9 \pm 3.249836 * (3/\sqrt{10})$

(4.82 10.98)

Group 8

Scientists conducted a study to examine the effect of light at night on body mass gain in mice. They randomly assigned the mice to one of three groups: a control group that had a normal light/dark cycle within darkness at night, a dim light group that exposed the mice to a dim light all night long (instead of complete darkness), and a bright light group, where they exposed mice to a bright light all night long (instead of complete darkness). The body mass gain, in grams, was recorded after eight weeks.

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1. In the dim light group, there were 10 mice. After eight weeks, they gained an average of 7.9g with a standard deviation of 3.0g. Calculate a **95% confidence interval for the mean weight gain** of mice exposed to dim light at night. (Here, we are assuming that the mice in this group are representative of all mice.) Show your work. If you use R, tell what commands you use.

qt(p=0.975, df=9)

→ output: 2.262157 (t*)

$\bar{x} \pm t^* \left(\text{sample standard error} / \sqrt{\text{sample size}} \right)$

95% CI: **7.9 ± 2.262157*(3.0/sqrt(10))**

[5.753929, 10.04607]

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

`qt(p=0.995, df=9)`

$$1 - 0.005 = 0.995 \leftarrow 99\%$$

$$1 - 0.025 = 0.975 \leftarrow 95\%$$

$$1 - 0.1 = 0.9 \leftarrow 80\%$$

$$t^* = 3.249836$$

$$99\% \text{ CI: } 7.9 \pm 3.249836 \cdot (3.0/\sqrt{10})$$

$$99\% \text{ CI: } [4.816935, 10.98307]$$

Group 9

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Sample size: $n = 10$

R code: `qt(0.975, df = 9)`

Sample mean: $\bar{x} = 7.9\text{g}$

Sample standard deviation = 3.0g

$\bar{x} \pm t^*(S/\sqrt{n})$

$t^* = 2.262$

Calculate SE = $3.0/\sqrt{10} = 0.9487$

Calculate ME = $t^* \times \text{SE} = 2.262 \times 0.9487 = 2.145$

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

Find the new t

R code: $qt(0.995, df = 9) = 3.250$

SE is the same = 0.9487

Calculate ME = $3.250 * 0.9487 = 3.083$

Confidence interval = lowerBound = 4.817, upperBound = 10.983

Ans (4.82, 10.98)

Group 10

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$$qt(0.975, df = 9) = 2.262157$$

$$7.9 + 2.262157 * (3.0/\sqrt{10}) = 10.04607$$

$$7.9 - 2.262157 * (3.0/\sqrt{10}) = 5.753929$$

2. Now calculate a 99% confidence interval for mean weight gain of mice exposed to dim light at night..

$$qt(0.995, df = 9) = 3.249836$$

$$7.9 + 3.249836 * (3.0/\sqrt{10}) = 10.98307$$

$$7.9 - 3.249836 * (3.0/\sqrt{10}) = 4.816935$$